



HDTV by Hitachi

similar concept to digital photos and how dpi determines print quality.

In the conventional TV system, waves are scanned at 480i. This means that at a given point of time four hundred and eighty (480) lines of pictures are sent by the broadcaster to the television. The lines alternate between odd and even in a succession that is too fast for the human eye.

HDTV sends around a thousand eighty (1080) or seven hundred and twenty (720) lines, which is more than double the conventional TV. By almost doubling the amount of lines in combination with the type of scan, HD

essentially nearly doubles the quality of picture.

R. Zutshi, deputy managing director of Samsung India, says that high-definition TVs offer the best TV viewing possible. HD format contains more and finer detail than other formats, so images can have almost lifelike clarity. The picture quality can be stunning, especially on a large, wide-screen set. In technical terms, HD images have higher definition, meaning more picture elements (lines or pixels) make up each image.

HDTV provides wide-screen picture quality with compact disk (CD) sound quality. It has over twice the sharpness and clarity of analogue TV broadcasts together with a far superior colour resolution. With HDTV, problems such as double images from ghosting or multi-path, snow caused from a weak signal and picture sparkles from impulse noise are a thing of the past. These problems often seen on a conventional television broadcast just do not occur on high-definition TV. Also, the sound for conventional TV is only stereo.

However, currently, HDTV broadcast does not take place in India, and the usage of HD resolution is limited to DVDs or other media only, so a high-

definition TV is effective if it is used for viewing DVD titles, etc.

What all is required for HDTV reception?

You may ask "Do I need a special TV to be able to see HDTV?" The answer is that only high-definition televisions can display HD content in the HD resolution. Enhanced-definition televisions can display HD content in a 480p resolution, which is DVD quality. All other televisions that are analogue of standard digital will not be able to display HD content in an HD resolution because they lack the technology to do so.

As is clear, anyone who owns a high-definition television can get high-definition content. One has three options: over-the-air signals, cable or satellite. Over-the-air high-definition signals are those that a typical rooftop antenna would receive. The signals are digital and encoded in HD. Over-the-air signals are free to receive. The only cost out of pocket would be for the equipment needed to receive them. To receive HD programming from a cable or satellite provider, you would need to subscribe to their HD package if they have any. This subscription may not be free. The provider might require

Sources of HDTV Signals

You can receive HD signals from the following sources:

1. Over the air with an antenna. Over-the-air signals are free to receive. However, to receive these signals, you need an HD tuner. Latest HDTVs have an HD tuner built into them. For HDTV televisions without a built-in HD tuner, a separate set-top HD tuner box can be rented from a cable or satellite company, or purchased.
2. Cable or satellite service. To receive HD programming from a cable or satellite provider, you would need to subscribe to their HD package if they have any. This subscription may not be free. The provider might require a minimum length of service. You require a set-top box issued by the cable company, to receive HD content. DishTV and Reliance Communications (BIG TV) are among the satellite TV providers in India who plan to offer HDTV-ready set-top boxes.
3. Blu-ray disk can store over nine hours of high-definition (HD) video on a 50GB disk.
4. A DVD-R can also store up to three hours of HD video content, readable by Blu-ray player, PlayStation 3 video game console or Blu-ray drive installed in a PC, depending on encoder settings.
5. Video game consoles. The latest generation of game consoles is more HD-friendly. While Xbox 360 and PlayStation 3 let you enjoy HD gaming, the Wii will not. The PlayStation 3 also doubles as a Blu-ray movie player.
6. If your computer graphics card supports the standard SVGA resolution, it can be used to output video to an HDTV's "PC input" jack.
7. The DVI or HDMI interfaces of newer computer graphics cards can be used to output video to an HDTV.